NPTEL ONLINE CERTIFICATION COURSES



Course Name: Software Project Management

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TOPIC: Contract management



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CONCEPTS COVERED

- Introduction to managing contracts
- Types of contracts



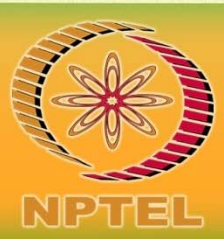
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Introduction

- Many organizations choose to obtain their software externally. Given their limited capability for developing new and reliable software, this seems sensible.
- These organizations think that it is more cost-effective to employ outside software developers for new development while a reduced group of in-house software development staff maintain and support existing systems.





Introduction cont...

- The buying of goods and services, rather than ‘doing it yourself’, is attractive when money is available but other, less flexible, types of resource, especially staff time, are in short supply.
- It is essential that customer organizations should find time to clarify their exact requirements at the beginning, and to ensure that the goods and services delivered are satisfactory.





Introduction cont...

- Potential suppliers are likely to be more accommodating before any contract is signed than afterwards – especially if the contract is for a fixed price.
- Thus, as much forethought and planning are needed with an acquisition project as with internal development.





Types of contracts

- *Bespoke system* - created specially for one customer
- *Off-the-shelf package* - bought ‘as is’ (also known as shrink-wrapped software)
- *Customized off-the-shelf* (COTS) software - a core system is customized to meet needs of the client

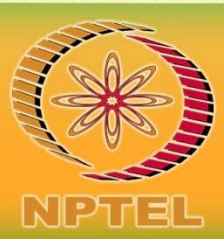




Types of contracts cont …

- Where an equipment is purchased, it is referred to as a contract for the supply of goods.
- With the supply of software this may be regarded as supplying a service (i.e. to develop the software) or the granting of a license (i.e. permission) to use the software which remains in the ownership of the supplier .





Types of contract cont …

Another way of classification based on the calculation of payment to suppliers is as follows:

- fixed price contracts
- time and materials contracts
- fixed price per delivered unit

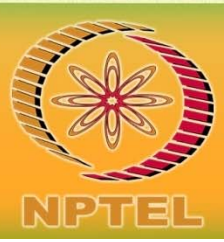




Fixed price contracts

- In this case, the price is fixed when the contract is signed.
- The customer knows that, if there are no changes in the contract terms, this is the price they will pay on completion.
- The customer's requirement has to be fixed at the outset, i.e. the detailed requirements analysis have already been made.
- Once the development is under way, the customer cannot change their requirements without renegotiating the price of the contract.





Fixed price contracts cont …

Advantages to customer

- known expenditure
- supplier motivated to be cost-effective





Fixed price contracts cont …

Disadvantages

- supplier will increase price to meet contingencies/absorb risks
- difficult to modify requirements
- cost of changes likely to be higher
- threat to system quality





Time and materials contracts

- In this case, the customer is charged at a fixed rate per unit of effort, for example, per staff-hour or per person-month.
- The supplier may provide an initial estimate of the cost based on their current understanding of the customer's requirements, but this is not the basis for the final payment.
- The supplier usually sends invoices to the customer for the work done at regular intervals, say monthly / quarterly.





Time and materials contacts cont …

Advantages to customer

- easy to change requirements
- lack of price pressure can assist product quality





Time and materials

Disadvantages

- Customer liability - the customer absorbs all the risk associated with poorly defined or changing requirements
- Lack of incentive for supplier to be cost-effective

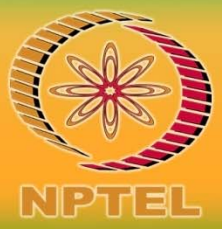




Fixed price per unit delivered contracts

- This is often associated with function point (FP) counting.
- The size of the system to be delivered is calculated or estimated at the outset of the project.
- The size could be estimated in lines of code, but FPs can be more easily derived from requirements documents.
- A price per unit is also quoted.
- The final price is then the unit price multiplied by the number of units.

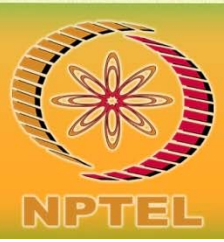




Fixed price per unit delivered contracts cont …

FP Count	Design cost/FP (in Rs.)	Coding cost/FP (in Rs.)	Total cost/FP (in Rs.)
upto 2000	200	700	900
2001-2500	240	760	1000
2501-3000	260	780	1040
3001-3500	270	800	1070
3501-4000	280	820	1100
4001-4500	300	840	1140
4501-5000	330	870	1200





Fixed price per unit delivered contracts cont …

- The company that produced this table in fact charges a higher fee per FP for larger systems.
- For example, a system to be implemented contains 2700 FPs.
- The overall charge would be 2000 X Rs 900 + 500 x Rs 1,000 + 200 x Rs 1,040 = Rs 25, 08,000.





Fixed price per unit delivered contracts cont …

- The scope of the application can grow during development.
- It would be unrealistic for a contractor to be asked to quote a single price for all the stages of a development project: how can they estimate the construction effort needed when the requirements are not yet established?
- In this case, one approach would be to negotiate a series of contracts, each covering a different stage of system development.





Fixed price per unit delivered contracts cont …

Advantages for customer

- customer understanding of how price is calculated
- comparability between different pricing schedules
- emerging functionality can be accounted for
- supplier incentive to be cost-effective





Fixed price per unit delivered contracts cont …

Disadvantages

- difficulties with software size measurement - may need independent FP counter
- changing (as opposed to new) requirements: how do you charge?





Types of software contracts cont …

Another way of classification based on the approach that is used in contractor selection is as follows:

- Open tendering process
- Restricted tendering process
- Negotiated procedure





The open tendering process

- Any supplier can bid in response to the *invitation to tender (ITT) / request for proposal (RFP)*
- All tenders must be evaluated in the same way
- Government bodies may have to do this by local/international law
- With a major project this evaluation process can be time consuming and expensive.
- Where the client is a public body, an open tendering process may be compulsory.





Restricted tendering process

- In this case, bids are received only from those suppliers who have been specifically invited
- Unlike, the open tendering process, at any time, the customer can reduce the number of suppliers being considered at any stage

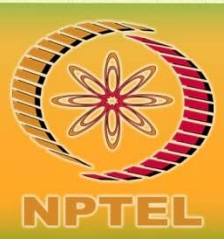
Negotiated procedure / Single tendering process

- Open or restricted tendering process may not be suitable in some particular sets of circumstances.
- For example, suppose, there is a fire that destroys some ICT equipment. The key concern here may be to get replacement equipment up and running as quickly as possible, and there may not be sufficient time to embark on a lengthy tendering process.

Negotiated procedure / Single tendering process cont …

- Another situation might be where a new software application had been successfully built by an outside supplier, but some extensions are required to the system. As the original supplier has staff familiar with the existing system, it might be inconvenient to approach other potential suppliers via a full tendering process.
- In these cases, an approach to a single supplier may be justified, i.e. negotiate with one supplier.





Negotiated procedure / Single tendering process cont …

- However, approaching a single, supplier could expose the customer to charges of favouritism.
- So, it should only be done with a proper justification.

Summary

- Presented an introduction to managing contracts.
- Discussed the different classifications of contracts along with their advantages and disadvantages.
- Also discussed which type of contract is best suitable in which cases.





References :

1. B. Hughes, M. Cotterell, R. Mall, *Software Project Management*, Sixth Edition, McGraw Hill Education (India) Pvt. Ltd., 2018.






Thank you
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